



Kresling-Based Salp Swimmer: Low Energy Aquatic Propulsion

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Princeton University, CEE 345: Origami Engineering



Introduction

Objective The salp (*salpidae*) is a gelatinous marine invertebrate related to jellyfish. They move through the water using a propulsion system where they contract and relax muscle bands that ring their bodies. Inspired by these creatures, our project aims to replicate this kind of movement with the bistable snapping of Kresling tubes to create an efficient model water propulsion system for robotic applications.

Inspiration: The Salp

Biological Inspiration

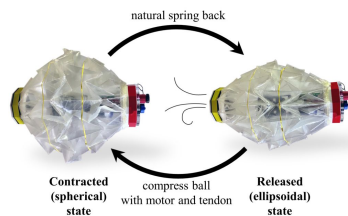
The goal was to build a bioinspired swimmer inspired by the highly efficient swimming gait of *Salpa fusiformis*.



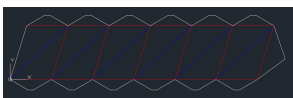
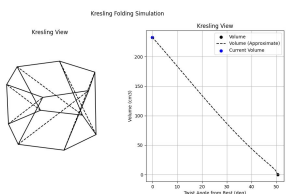
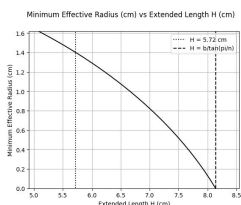
Early Ideas:

Sung Lab salp:

<https://sung.seas.upenn.edu/research/bio-inspired-soft-underwater-robot-that-swims-via-jet-propulsion/>



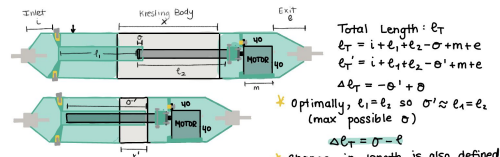
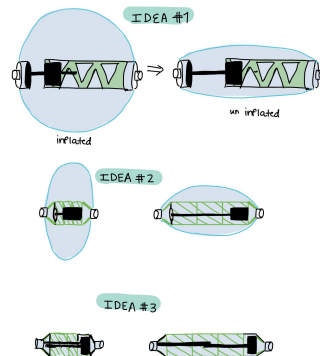
Theory and Analysis



Design: The Kresling

Methods

The biggest benefit of the bistable Kresling design is to efficiently convert the rotational motion of a motor into a large jump of linear contraction, displacing a large volume of water with a small amount of actuation.



Volume change (rough)

$\Delta V = \left(\frac{\pi}{2}\right)^2 \pi (x - x')$

assuming flat foldability:

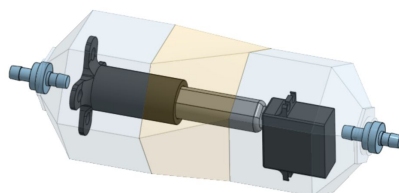
$\Delta V = \frac{1}{2} h^2 \pi x$

$\Delta V = \frac{1}{2} h^2 \pi (\ell - \theta)$

Mechanism

Motor-Driven Shaft Rotation

A servo motor turns a D-shaft, which rotates the outer shaft, turning one end of the Kresling. The Kresling turns rotational into linear motion.



Manufacturing & Assembly

Kresling

Initially, we attempted to use a 3D printed mold with silicone, but the walls of the structure were too flexible. We switched to a PET plastic chassis instead.



Manufacturing

We used epoxy to seal the one-way valves to each shell at the end of the kresling. We used the software-generated dxf files to lasercut the kresling out of PET plastic. To actuate the kresling we mounted internally a waterproof servo motor, which has the capacity to rotate. When the kresling is rotated, it translates the rotational motion into axial motion, resulting in the kresling compressing and propelling the salp forwards.

Future Steps

Future Steps

Future iterations will include better manufacturing, a more watertight Kresling, and electronics integrated into the body of the robot. The interface between the two d shafts tended to bind, so a better shaft rotation mechanism should be used in the future.



Additionally, the robots could be designed such that they can chain together, an emergent behavior seen in real salps in the wild, creating a much larger robot capable of multidirectional swimming

Acknowledgements

This work is supported by Princeton University, the CEE Department, Professor Paulino, Tuo Zhao, and Kevin Liu.